Hacettepe University - Computer Engineering
CMP720 Embedded System Design - Spring 2026

Final Project Topic Proposal Template

Group Members

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Project Topic Title

EdgeRAG: Offline Retrieval-Augmented Generation on Resource-Constrained Simulated ARM Architectures

Project Track

☒ **Implementation Track**

☐ **Optimization Track**

Abstract

Abstract

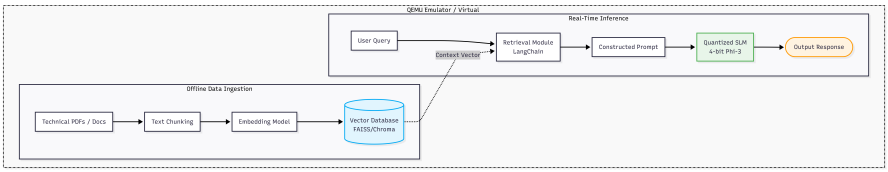
Deploying Large Language Models (LLMs) and intelligent reasoning systems typically relies on high-bandwidth cloud infrastructure. However, in embedded environments such as industrial IoT, remote sensors, and air-gapped secure facilities, continuous internet access is either impossible or poses significant security risks. The problem tackled in this project is the inability to run robust question-answering systems natively on resource-constrained embedded devices without relying on external servers.

To solve this, we propose an offline Retrieval-Augmented Generation (RAG) pipeline designed specifically for an edge environment. The methodology involves utilizing a lightweight vector database (such as FAISS or ChromaDB) and an optimized, heavily quantized Small Language Model (SLM), like a 4-bit GGUF TinyLlama or Phi-3, managed by a minimal LangChain architecture.

The high-level system architecture will be emulated using QEMU to simulate an ARM-based Single Board Computer (SBC) with artificially constrained resources (e.g., 2GB-4GB RAM limits and restricted CPU cycles). The system will consist of an ingestion module to parse documents locally, an embedding module to store vector representations, and a retrieval-generation module that answers natural language queries strictly operating on the emulated edge hardware.

The project will be executed through a series of structured milestones over an estimated 9 to 10-week period. The initial step involves establishing the QEMU ARM Linux simulation and enforcing hardware constraints. Following this, the quantized SLM will be deployed to evaluate basic CPU inference capabilities. The subsequent phase integrates the LangChain pipeline and local vector database for offline document ingestion and retrieval. The final steps are dedicated to rigorous system profiling—measuring critical embedded metrics such as memory footprint, CPU utilization, and inference latency—followed by comprehensive debugging, result analysis, and report generation.

System Pipeline Diagram:



Motivation / Societal Impact

Motivation and Societal Impact

The proliferation of AI has democratized information retrieval, but it has simultaneously created a strict dependency on cloud computing. This dependency marginalizes environments where internet connectivity is unstable, expensive, or strictly prohibited. By successfully migrating a functional RAG pipeline to a resource-constrained edge device, this project addresses critical issues regarding data privacy, system reliability, and accessibility.

From a societal impact perspective, the ability to run "EdgeRAG" ensures complete data privacy, which is essential for medical devices, defense systems, and proprietary industrial manufacturing. Since the data never leaves the device, the risk of interception or cloud-based data breaches is effectively reduced to zero. Furthermore, offline AI improves sustainability by drastically reducing the continuous energy costs and carbon footprint associated with transmitting massive amounts of data back and forth to server farms.

The primary beneficiaries of this research include field engineers working in remote locations without network access, privacy-focused consumers, and industries relying on air-gapped networks. We will measure the impact and success of this project quantitatively by profiling the system's performance within the emulator. Key metrics will include the maximum RAM consumption during vector retrieval versus text generation, the latency measured in tokens-per-second (t/s) under constrained CPU limits, and the success rate of querying large documents without causing system out-of-memory (OOM) failures.

Related Work / References

Recent advancements highlight the critical shift towards Edge AI [1, 2, 3, 4].

References

[1] J. W. Valvano, *Embedded Systems: Real-Time Interfacing to Arm Cortex-M Microcontrollers*. CreateSpace Independent Publishing Platform, 3rd ed., 2017.

[2] R. Sharma and K. Patel, "Rtos-based optimization techniques for resource-constrained embedded ai systems," *IEEE Access*, vol. 8, pp. 205123–205138, 2020.

[3] Espressif Systems, "ESP32 Series Datasheet." Technical Datasheet, 2023. Accessed: 2026-03-08.

[4] J. Lin and Y. Chen, "On-device llm inference and retrieval-augmented generation for embedded systems," *Journal of Edge Computing*, vol. 4, pp. 112–128, 2023.